0.1 Elementary cardinality

Definition 0.1.1. The sets A, B are **equinumerous** (or **have the same cardinality**) if there exists a bijection $f : A \to B$, we write this as $A \approx B$.

Definition 0.1.2. A set A is **finite** (resp. **countable infinite**) if it is equinumerous with some natural number (resp. $\mathbb{N}$), and just **countable** if it is either of those.

Recall the following basic facts about countable sets from IUM:

- Every subset of a countable set is countable;
- a set A is countable iff there exists an injective function $f : A \to \mathbb{N}$;
- if A and B are countable then $A \times B$ is countable;
- if $A_0, A_1, \dots$ are all countable, then their union $\bigcup_n A_n$ is also countable, but this uses the axiom of choice.

Theorem 0.1.3 (Cantor)**.** For all sets X, there is no surjective function $f : X \to \mathcal{P}(X)$.

Proof. Suppose, for a contradiction, there exists such an f, then consider the set

$$Y = \{y \in X \mid y \notin f(y)\}$$

As f is surjective, there exists a set $z \in X$ with $f(z) = Y$, but now if we assume either $z \in Y$ or $z \notin Y$ we get a contradiction. □

Definition 0.1.4. For sets A, B, write $|A| \leq |B|$ if there exists an injective function $f : A \to B$.

We will not develop the concept of cardinality $|A|$ for a while, so for now this symbol will only appear as a relation.

Theorem 0.1.5 (Cantor-Schröder-Bernstein)**.** Suppose A, B are sets and there exists injective functions $f : A \to B$ and $g : B \to A$, then $A \approx B$.

Proof. First consider the sequence of sets $A_0 := A \setminus g(B)$ and for all $n > 0$, $A_n := (g \circ f)(A_{n-1})$. Then write $A^* := \bigcup_n A_n$ and $B^* = f(A^*)$, these clearly satisfy $g(B^*) = (g \circ f)(A^*) \subseteq A^*$.

I now claim $g(B \setminus B^*) = A \setminus A^*$. For all $a \in A \setminus A^*$, as $a \notin A_0$ there must exist a $b \in B$ with $g(b) = a$. If $b \in B^* = f(A^*)$, then $a = g(b) \in (g \circ f)(A^*) \subseteq A^*$ a contraduction, therefore $A \setminus A^* \subseteq g(B \setminus B^*)$. Now for all $b \in B$, if $g(b) \in A^*$, then as $g(b) \notin A_0$ there must exist some $n > 0$ with $g(b) \in A_n$ and thus $g(b) = (g \circ f)(a)$ for some $a \in A_{n-1}$, so $b = f(a)$ for some $a \in A^*$ and thus $b \in f(A^*) = B^*$.

g now gives a bijection $B \setminus B^* \to A \setminus A^*$, and as f is a bijection $A^* \to B^*$ we can form the bijection

$$k(a) = \begin{cases} f(a) & \text{if } a \in A^* \\ g^{-1}(a) & \text{if } a \in A \setminus A^* \end{cases}$$ □

0.2 Zermelo-Fraenkel axioms

We will express these axioms in a first-order language with equality, which we call $\mathcal{L}_{\in}^{=}$, with a single extra 2-ary relation symbol $\in$.

Axiom 1 (Extensionality)**.** $(\forall x)(\forall y)((x = y) \leftrightarrow (\forall z)((z \in x) \leftrightarrow (z \in y)))$

Axiom 2 (Empty set)**.** $(\exists x)(\forall y)(y \notin x)$

Axiom 3 (Pairing)**.** $(\forall x)(\forall y)(\exists z)(\forall w)((w \in z) \leftrightarrow ((w = x) \vee (w = y)))$

By extensionality we know we can write $\emptyset$ to mean the unique empty set in ZF. We can also start to form the natrual numbers $0 = \emptyset$, $1 = \{\emptyset, \emptyset\} = \{\emptyset\}$, and $2 = \{\emptyset, \{\emptyset\}\}$, but we can't yet go any further.

Axiom 4 (Union)**.** $(\forall A)(\exists B)(\forall x)((x \in B) \leftrightarrow (\exists z)((z \in A) \wedge (x \in z)))$

So if we take $A = \{x, y\}$ then $B = x \cup y$, so we can now form $3 = \{0, 1, 2\} = \{0, 1\} \cup \{2\}$, and thus we can form all the natrual numbers.

We will want to adopt the abreviation $A \subseteq B$ to mean $(\forall x)(x \in B \to x \in A)$.

Axiom 5 (Powerset)**.** $(\forall A)(\exists B)(\forall z)((z \in B) \leftrightarrow (z \subseteq A))$

Axiom 6 (Specification)**.** For all $\mathcal{L}_{\in}^{=}$-formulas $P(x, y_1, \ldots, y_r)$, for some finite number of variables r, there exists the axiom $(\forall A)(\forall y_1) \ldots (\forall y_r)(\exists B)((x \in B) \leftrightarrow ((x \in A) \wedge P(x, y_1, \ldots, y_r)))$.

This lets us form the intersection, let C be a nonempty set with $A \in C$, then

$$\bigcap C := \{x \in A \mid (\forall z)((z \in C) \to (x \in z))$$

and for sets A, B, we can form the product

$$A \times B := \{w \in \mathcal{P}(\mathcal{P}(A \cup B)) \mid (\exists a)(\exists b)((a \in A) \wedge (b \in B) \wedge (w = \{\{a\}, \{a, b\}\}))\}$$

and the set of functions

$$B^A := \{w \in \mathcal{P}(A \times B) \mid (\forall a)((\exists x)(a \in x \wedge x \in w) \wedge (\forall y)(\forall z)((a \in y \wedge a \in z) \to y = z))\}$$

Definition 0.2.1. For all sets a, define the **successor** of a to be the set $a^\dagger := a \cup \{a\}$. A set A is **inductive** if $(\emptyset \in A) \wedge (\forall x)((x \in A) \to (x^\dagger \in A))$.

Axiom 7 (Infinity)**.** There exists an inductive set.

Definition 0.2.2. We can use specification to form the set of inductive sets, and take their intersection, call this ω.

Proposition 0.2.3. $\mathbb{N}$ is an inductive set, and is B is an inductive set, then $\mathbb{N} \subseteq B$. If $A \subseteq \mathbb{N}$ is inductive, then $A = \mathbb{N}$

This is proof by induction, if P is an $\mathcal{L}_{\in}^{=}$-formula and we let $A = \{x \in \mathbb{N} \mid P(x)\}$, then if we prove a base case and an inductive step, we get $A = \mathbb{N}$.

0.3 Well orderings

Definition 0.3.1. A linear ordering $(A; \leq)$ is a **well-ordering** if every nonempty subset of A has a least element.

Definition 0.3.2. If two linear orders $\mathcal{A}_1 = (A_1; \leq_1)$ and $\mathcal{A}_2 = (A_2; \leq_2)$ are isomorphic we call them **similar** and write $\mathcal{A}_1 \simeq \mathcal{A}_2$. Furthermore, if $\beta : A_1 \to A_2$ is injective and $\forall a, b \in A_1$ we have $a \leq_1 b \implies \beta(a) \leq_2 \beta(b)$, we say β is **order-preserving**.

Definition 0.3.3. Given $\mathcal{A}_1, \mathcal{A}_2$ as above the **reverse lexicographic product** is the linear order $\mathcal{A}_1 \times \mathcal{A}_2 := (A_1 \times A_2, \leq)$ with the ordering $(a_1, a_2) \leq (b_1, b_2)$ iff either $a_2 \leq_2 b_2$ or $a_2 = b_2$ and $a_1 \leq_1 b_1$. Their **sum** is the linear order $(A_1 \sqcup A_2; \leq)$ where $\leq$ is the union of $\leq_1$, $\leq_2$, and $a_1 \leq a_2$ for all $a_1 \in A_1$ and $a_2 \in A_2$.

The key example is $\{0, 1\} \times \mathbb{N} \simeq \mathbb{N} \not\simeq \mathbb{N} + \mathbb{N} \simeq \mathbb{N} \times \{0, 1\} \not\simeq \mathbb{N} \times \mathbb{N}$. The proof that the sum and product actually satisfy the linear ordering axioms is left as an exercise.

Lemma 0.3.4. If $\mathcal{A}_1$ and $\mathcal{A}_2$ are well-orderings, then so are $\mathcal{A}_1 + \mathcal{A}_2$ and $\mathcal{A}_1 \times \mathcal{A}_2$.

0.4 Ordinals

Definition 0.4.1. A set X is called **transitive** if every element of X is also a subset of X.

A set α is called an **ordinal** if α is a transitive set and the relation $<$ on α given by $x < y \iff x \in y$ is a strict well ordering.

Lemma 0.4.2. If α is an ordinal, then so is $\alpha^\dagger$.

Proof. For transitivity, if $\beta \in \alpha^\dagger$ then either $\beta \in \alpha$ or $\beta = \alpha$, either way $\beta \subset \alpha^\dagger$. As the ordering $\in$ makes α the maximal element of $\alpha^\dagger$, this ordering is still strict, and by definition $\alpha \notin \alpha$, so the ordering remains strict. □

So we can do induction to see that all the natural numbers are ordinals.

Lemma 0.4.3. If α is a ordinal and $\beta \in \alpha$, then β is an ordinal.

Proof. If $y \in x \in \beta \in \alpha$, then as $\in$ is an ordering, so transitive, we get $y \in \beta$, so β is transitive and inherets the ordering from α, thus is an ordinal. □

Lemma 0.4.4. If α and β are ordinals, and $\alpha \subset \beta$, then $\alpha \in \beta$.

Proof. As $\beta \setminus \alpha$ is nonemtpy, we may choose a minimal element γ, I now claim $\gamma = \alpha$.

For all $x \in \gamma$, we have $x \in \beta$ and so by minimality $x \in \alpha$, thus $\gamma \subseteq \alpha$. Now suppose there exists some $y \in \alpha \setminus \gamma$ then in the total ordering on β we must have either $y = \gamma$ or $\gamma \in y$, either way $\gamma \in \alpha$ by transitivity. □

Definition 0.4.5. If α and β are ordinals, write $\alpha < \beta$ for $\alpha \in \beta$.

Proposition 0.4.6. Suppose α, β, and γ are oridnals, then the following all hold:

1. if $\alpha < \beta$ and β, γ, then $\alpha < \gamma$;
2. if $\alpha \leq \beta$ and $\beta \leq \alpha$, then $\alpha = \beta$;
3. exactly one of $\alpha < \beta$, $\alpha = \beta$, or $\alpha > \beta$ holds;
4. if X is a nonemtpy set of ordinals, then X has a least element.

Proof. (1) is just by transitivity of γ.

(2) must hold, otherwise we would have $\alpha \in \beta \in \alpha$ a contradiction.

(3) We know at most one of these hold, and from the definition we can see $\alpha \cap \beta$ is also an ordinal. If neither hold, $\alpha \not\subseteq \beta$ and $\beta \not\subseteq \alpha$, then $\alpha \cap \beta \subset \alpha$ hence $\alpha \cap \beta \in \alpha$, by doing the same with β we get $\alpha \cap \beta \in \alpha \cap \beta$, which is a contradiction.

(4) Take $\delta \in X$, then the least element of X is the least element of $\delta \cap X \subseteq \delta$. □

Corollary 0.4.7. If X is a nonempty set of ordinals, then $\bigcup X$ is an ordinal.

Proof. If $y \in z \in \bigcup X$, then there exists some $x \in X$ with $z \in x$, so $y \in x$, and thus $y \in \bigcup X$. □

Corollary 0.4.8. ω is an ordinal.

Definition 0.4.9. If $(A; \leq)$ is a well-ordering, then we call $X \subset A$ an **initial segment** of A if whenever $y < x \in X$, then $y \in X$, X is proper if $X \neq A$.

Lemma 0.4.10. If $(A; \leq)$ is a well order, with X a proper initial segment, then there exists $x \in A$ such that $X = \{a \in A \mid a < x\}$.

Proof. Let x be the minimal element of $A \setminus X$, then for all $y < x$ we have $y \in X$. If $z \in X$, then $z \neq x$, so if $z > x$ we must have $x \in X$, a contradiction. □

Lemma 0.4.11. If $(A; \leq)$ is a well-ordering, $f : A \to A$ is order-preserving, and $f(A)$ is an initial segment of A, then $f(a) = a$ for all $a \in A$.

Proof. Suppose this was not the case, then consider x, the least element of the set $\{y \in A \mid f(y) \neq y\}$, then f is the identity on the initial segment $A[x]$, and hence as f is injective and $f(x) \neq x$, we must have $f(x) > x$. But now $x \notin f[A]$, even though x is order preserving, a contradiction. □

Corollary 0.4.12. If $\alpha \neq \beta$ are ordinals, then $\alpha \not\simeq \beta$.

For the next proof we need the next ZF axiom.

Axiom 8 (Replacement)**.** Suppose $F(x, y, z_1, \ldots, z_r)$ is a property of sets expressible in $\mathcal{L}_{\in}^{=}$ such that, whenever $s_1, \ldots, s_r$ are sets, for every set a there exists a unique set b such that $F(a, b, s_1, \ldots, s_r)$ holds. Then for any set A, we can form the set

$$B = \{b \mid (\exists a)((a \in A) \wedge F(a, b, s_1, \ldots, s_r))\}$$

replacing each $a \in A$ with the unique b such that $F(a, b, s_1, \ldots, s_r)$ holds.

Theorem 0.4.13. If $(A; \leq)$ is a well-ordering, then there exists a unique ordinal α which is similar to $(A; \leq)$.

Proof. First, uniqueness is a exactly the previous corollory, as if α and β are both similra to $(A, \leq)$, then they are similar to each other, and hence equal.

For existence, consider the set $X = \{x \in A \mid A[x] \text{ is similar to an ordinal}\}$, then for each x there is a unique ordinal α_x such that $\alpha_x \simeq A[x]$, so we can use replacement to form the set $S = \{\alpha_x \mid x \in X\}$. As S is a set of ordinals, to show it is an ordinal itself, it suffices to show transitivity. If we have $\beta \in \alpha_x \in S$, with $\pi : A[x] \to \alpha_x$ the similarity, then we let $y = \pi^{-1}(\beta)$ which makes π a similar $A[y] \to \beta$, so $\beta = \alpha_y \in S$. This in fact says X is an initial segment of A. Suppose it is proper, then we must have $X = A[z]$ for some $z \in A$, but then $z \in X = A[z]$ a contradiction, so $X = A$ and the map $x \mapsto \alpha_x$ is a similarity between X and S. □

0.5 Transfinite induction and recursion

Theorem 0.5.1 (Transfinite induction)**.** Suppose $P(x)$ is a property of sets such that for all ordinals α, if $P(\beta)$ holds for all ordinals $\beta < \alpha$, then $P(\alpha)$ holds; then $P(\gamma)$ holds for all ordinals γ.

Proof. If $\alpha = 0$, then $P(\beta)$ holds vacuously, so $P(0)$ holds. Now consider the set of ordinals γ for which P doesn't hold, by assumption this can't have a minimum, and so must be empty. □

Theorem 0.5.2. For all infinite ordinals α, we have $\alpha \approx \alpha \times \alpha$.

Proof. We already know the result holds if α is any countable ordinal, so we may assume α is uncountable, now by transfinite induction we just have to assume for all $\omega \leq \beta < \alpha$, that $\beta \approx \beta \times \beta$ and show $\alpha \approx \alpha \times \alpha$. We may also assume that if $\beta < \alpha$, then $\beta \not\approx \alpha$, from which it follows $\beta^\dagger \not\approx \alpha$. Finally, by Cantor-Schröder-Bernstein we just need to find an injective function $\alpha \times \alpha \to \alpha$.

First, suppose we have a well-ordering $\leq$ of $A = \alpha \times \alpha$ such that for all $x \in A$, $|A[x]| < |\alpha|$, then we may find an ordinal γ with $\gamma \simeq (A; \leq)$, and call the similarity f, then for all $\eta \in \gamma$ as f is a similarity, it gives a bijection $\eta \to A[f(\eta)]$, so $\eta \approx A[f(\eta)]$ which has cardinality strictly less than α, so by trichotomy we have $\eta < \alpha$ equivalently $\eta \in \alpha$, so $\gamma \subseteq \alpha$ and we are done.

Now, we must produce such a well-ordering on A. For all $\lambda < \alpha$ consider the set $A_\lambda := \{(\theta, \zeta) \mid \max(\theta, \zeta) = \lambda\}$ then we define $\leq$ on A by $(\theta, \zeta) \leq (\theta', \zeta')$ iff $\max(\theta, \zeta) < \max(\theta', \zeta)$ or if $\max(\theta, \zeta) = \max(\theta', \zeta')$ and either $\zeta < \zeta'$ or $\zeta = \zeta'$ and $\theta \leq \theta'$, it is left as an exercise to show this is a well-ordering on A. Now for $x = (\theta, \zeta) \in A$, if we take $\lambda = \max(\theta, \zeta)$, then $A[x] \subseteq A_{\lambda^\dagger} \subset \lambda^\dagger \times \lambda^\dagger$, and by our initial assumptions, we know $\lambda^\dagger \times \lambda^\dagger \approx \lambda$ which has cardinality strictly less than α. □

Corollary 0.5.3. If $(A; \leq)$ is an infinite well-ordering, then $A \approx A \times A$.

Proof. We may find an ordinal α with $A \approx \alpha \approx \alpha \times \alpha \approx A \times A$. □

Corollary 0.5.4. Assuming the axiom of choice, if A is any infinite set, then $A \approx A \times A$.

Proof. We will see that the axiom of choice lets us make any set well-ordered. □

From the axioms of replacement, we have the notion of an operation F on sets, some property $F(x, y)$ of sets such that for each set a there is a unique set b such that $F(a, b)$ holds.

Theorem 0.5.5 (Transfinite recursion)**.** If F is an operation on sets, then there is an operation G such that for all ordinals α, $G(\alpha) = F(G \restriction \alpha)$, where we define $G \restriction \alpha : \alpha \to \{G(\beta) \mid \beta < \alpha\}$. If G' is another operation such that $G'(\alpha) = F(G' \restriction \alpha)$ for all ordinals α, then $G(\alpha) = G'(\alpha)$ for all oridnals α.

Proof. Non-examinable. □

We can use this to prove a version of the Lindenbaum lemma which doesn't assume $\mathcal{L}$ is countable.

Proposition 0.5.6. Suppose $\mathcal{L}$ is a first-order language whose alphabet of symbols A is well-ordered, then if Σ is a closed set of $\mathcal{L}$-formulas, we can find a complete consisten set $\Sigma^* \supseteq \Sigma$ of closed $\mathcal{L}$-formulas.

Proof. As A is well-ordered, we can also well order the set of finite sequences in A which we call S. Hence we get a well-ordering on any subset of S, in particular the set of closed $\mathcal{L}$-formulas. This set of closed $\mathcal{L}$-formulas is now similar to some ordinal λ and index our set of $\mathcal{L}$-formulas as $\{\phi_\alpha \mid \alpha < \lambda\}$.

Now, for each ordinal α, define the a consistem set $G(\alpha) \supseteq \Sigma$ of closed $\mathcal{L}$-formulas by transfinite induction as follows:

$$G(\alpha) = \begin{cases} \Sigma \cup \bigcup\limits_{\beta<\alpha} G(\beta) \cup \{\phi_\alpha\} & \text{if } \alpha < \lambda \text{ and } \Sigma \cup \bigcup\limits_{\beta<\alpha} G(\beta) \vdash \phi_\alpha \\ \Sigma \cup \bigcup\limits_{\beta<\alpha} G(\beta) \cup \{(\neg\phi_\alpha)\} & \text{if } \alpha < \lambda \text{ and } \Sigma \cup \bigcup\limits_{\beta<\alpha} G(\beta) \nvdash \phi_\alpha \\ \Sigma \cup \bigcup_{\beta<\lambda} G(\beta) & \text{if } \alpha \geq \lambda \end{cases}$$

now with the argument similar to the countable case, by with transfinite induction, we see for all $\beta < \alpha$, we have $G(\beta) \subseteq G(\alpha)$, and each $G(\alpha)$ is complete. Now if we set $\Sigma^* = G(\lambda)$, this is consistent and complete. □

We can use similar arguments to give generalisations of the other parts of the Model existence theorem, which will eventually give us a Completeness and Compactness theorem for any $\mathcal{L}$ with a well-ordered alphabet.

Axiom 9 (Regularity)**.** $(\forall x)((x \neq \emptyset) \to (\exists a)((a \in x) \wedge (a \cap x = \emptyset)))$

We have now stated all nine axioms of Zermelo-Fraenkel set theory.

0.6 Choice

Axiom 10 (Choice)**.** Suppose A is a set of nonempty sets, then there exists a function $f : A \to \bigcup A$, such that for all $a \in A$, we have $f(a) \in a$.

If X is a non-empty set and $A = \mathcal{P}(X) \setminus \{\emptyset\}$, then AC gives us a function $f : A \to X$, such that for each $\emptyset \neq Y \subseteq X$, we have $f(Y) \in Y$.

If my set comes with a well-ordering, then by taking $f(Y)$ to be the minimum element of Y, I don't need the axiom of choice, this sort of goes the other way.

Theorem 0.6.1 (Hartog's lemma)**.** For any set X, there exists an ordinal α such that there is no injective function $h : \alpha \to X$.

Proof. Consider the set $\mathcal{Y} = \{(Y; \leq_Y) \mid Y \subseteq A, \ \leq_Y \text{ is a well order on } Y\}$, then we use the axiom of replacement to form S by replacing each $(Y; \leq_Y)$ with an ordinal similar to it. Then S is in fact the set of ordinals which admit an injection into X, then let $\sigma = \bigcup S$, an ordinal, and now for all $\beta \in S$, we have $\sigma^\dagger > \beta$, so $\sigma^\dagger \notin S$. □

Theorem 0.6.2. Suppose X is a non-empty set, and $f : \mathcal{P}(X) \setminus \{\emptyset\} \to X$ is a choice function, then there is a well-ordering $\leq$ on X.

Proof. Let q be some set with $q \notin X$, and consider $\tilde{X} = X \cup \{q\}$, using transfinite recursion, there is an operation G by for all ordinals γ:

$$G(\gamma) = \begin{cases} f(X \setminus \{G(\beta) \mid \beta, \gamma\}) & \text{if } X \setminus \{G(\beta) \mid \beta < \gamma\} \neq \emptyset \\ q & \text{otherwise} \end{cases}$$

if $q \notin \operatorname{im}(G \restriction \gamma)$ for some γ, then $G \restriction \gamma$ is an injective function $\gamma \to X$, now by Hartog's lemma, there is some ordinal α with $G(\alpha) = q$, if we take the least such α, then $G \restriction \alpha : \alpha \to X$ is a bijection, and this defines a well-ordering on X. □

Axiom 11 (Well ordering principle)**.** For all sets A, there is a well-ordering on A.

Corollary 0.6.3 (ZF)**.** The well ordering principle is equivalent to the axiom of choice.

Proof. The axiom of choice gives a choice function of every set, which gives a well-ordering as above. If a set is well-ordered, then we may take the minimum element choice function we defined earlier. □

Corollary 0.6.4 (ZFC)**.** For all sets A, there exists an ordinal with $A \approx \alpha$.

Corollary 0.6.5 (ZFC)**.** For all sets A and B, either $|A| \leq |B|$ or $|B| \leq |A|$.

Corollary 0.6.6 (ZFC)**.** If A is any infinite set, then $A \times A \approx A$.

Lemma 0.6.7 (ZFC)**.** Suppose A and B are non-empty sets, then there is an injective function $A \to B$ iff there is a surjection $B \to A$.

0.7 Cardinals

Definition 0.7.1. An ordinal α is called a **cardinal** if for all $\beta < \alpha$, $\alpha \not\approx \beta$.

From the problem sheets we know all finite ordinals and ω are cardinals, if ω is an ifnite ordinal, then $\omega^\dagger$ isn't a cardinal, the set of countable ordinals is a cardinal, and every set A is equinumerous to a unique cardinal α.

Definition 0.7.2. The α above is called the **cardinality** of A, and we write $|A| = \alpha$.

So if we accept the axiom of choice, we get a much shorter proof of Cantor-Schröder-Berstein just by comparing cardinalities.

Definition 0.7.3. We define the sequence of **aleph** cardinals by transfinite recursion as follows. For all ordinals α, $\aleph_\alpha$ is a cardinal. $\aleph_0 = \omega$, and $\aleph_\alpha$ is the least cardinal which is greater than $\aleph_\beta$ for all ordinal $\beta < \alpha$. The existence of these is guaranteed by Cantor's theorem.

Definition 0.7.4 (Cardinal arithmetic)**.** If A and B are disjoint sets with cardinalities $|A| = \kappa$ and $|B| = \lambda$, then define $\kappa + \lambda = |A \cup B|$, $\kappa \cdot \lambda = |A \times B|$, and $\kappa^\lambda = \left|A^B\right|$.

These cardinalities don't depend on the choice of A and B.

Theorem 0.7.5. If $\kappa \leq \lambda$ are caridinals with λ infinite, then $\kappa + \lambda = \kappa \cdot \lambda = \lambda$. If $2 \leq \kappa \leq \lambda$ then $2^\lambda = \kappa^\lambda$.

Proof. For the first statement, as $\kappa \leq \lambda$, we have $\kappa \cdot \lambda = |\kappa \times \lambda| \leq |\lambda \times \lambda| = |\lambda| = \lambda$, and we can easily find an injection $\lambda \to \kappa \times \lambda$. We also have $\lambda \leq \kappa + \lambda \leq \lambda + \lambda \approx 2 \cdot \lambda \approx \lambda$. The second part is left as an exercise. $\square$

Theorem 0.7.6. Suppose A is an infinite set of cardinality λ, and each element of A has cardinality at most κ, then $|\bigcup A| \leq \lambda \cdot \kappa$.

Proof. For each $a \in A$, the set S_a of surjective functions $\kappa \to a$ is non-empty, so by AC there is a function $F : A \to \bigcup S_a$ with $F(a) \in S_a$. Let $h : \lambda \to A$ be a bijection, then define $G : \lambda \times \kappa \to \bigcup A$ by $G(\alpha, \beta) = F(h(\alpha))(\beta)$, this is surjective and thus $|\bigcup A| \leq \lambda \cdot \kappa$. $\square$

From this we can deduce that for any infinite set A, the set of finite sequences of elements of A has the same cardinality. And, if B is a $\mathbb{Q}$-basis for $\mathbb{R}$, then $|B| = |\mathbb{R}|$.

0.8 Zorn's lemma

Definition 0.8.1. A **chain** is a subset of a partially ordered set which is linearly ordered.

Axiom 12 (Zorn's lemma)**.** If $(A; \leq)$ is a non-emtpy partially ordered set, in which every chain has a nupper bound, then A has a maximal element.

Theorem 0.8.2. Assuming ZF, Zorn's lemma is equivalent to the axiom of choice.

Proof. (AC$\Rightarrow$ZL) Given a poset $(A; \leq)$ satisfying the hypothesis of Zorn's lemma, let $f : \mathcal{P}(A) \setminus \{\emptyset\} \to A$ be a choice function, and suppose, for a contradiction, that A has no maximal element. Let $C \subseteq A$ be a chain, then by assumption there exists some $y \in A$ with $C \leq y$, and so there exists some $z \in A$ with $z > y$, so $C \subsetneq C \cup \{z\}$ is another chain, and we now use transfinite recursion to define an operation G such that for all ordinals $\beta < \alpha$, $G(\alpha) \in A$ and $G(\beta) < G(\alpha)$. For this we let

$$G(\alpha) = f(\{z \in A \mid z > G(\beta) \text{ for all } \beta < \alpha\})$$

which gives for every α an injection $G \restriction \alpha : \alpha \to A$, which contradicts Hartog's lemma.

The other direction of the proof is left as an exercise. $\square$